# Picture World Maker 2.1 - User Manual

### Installing Picture World Maker - read this first

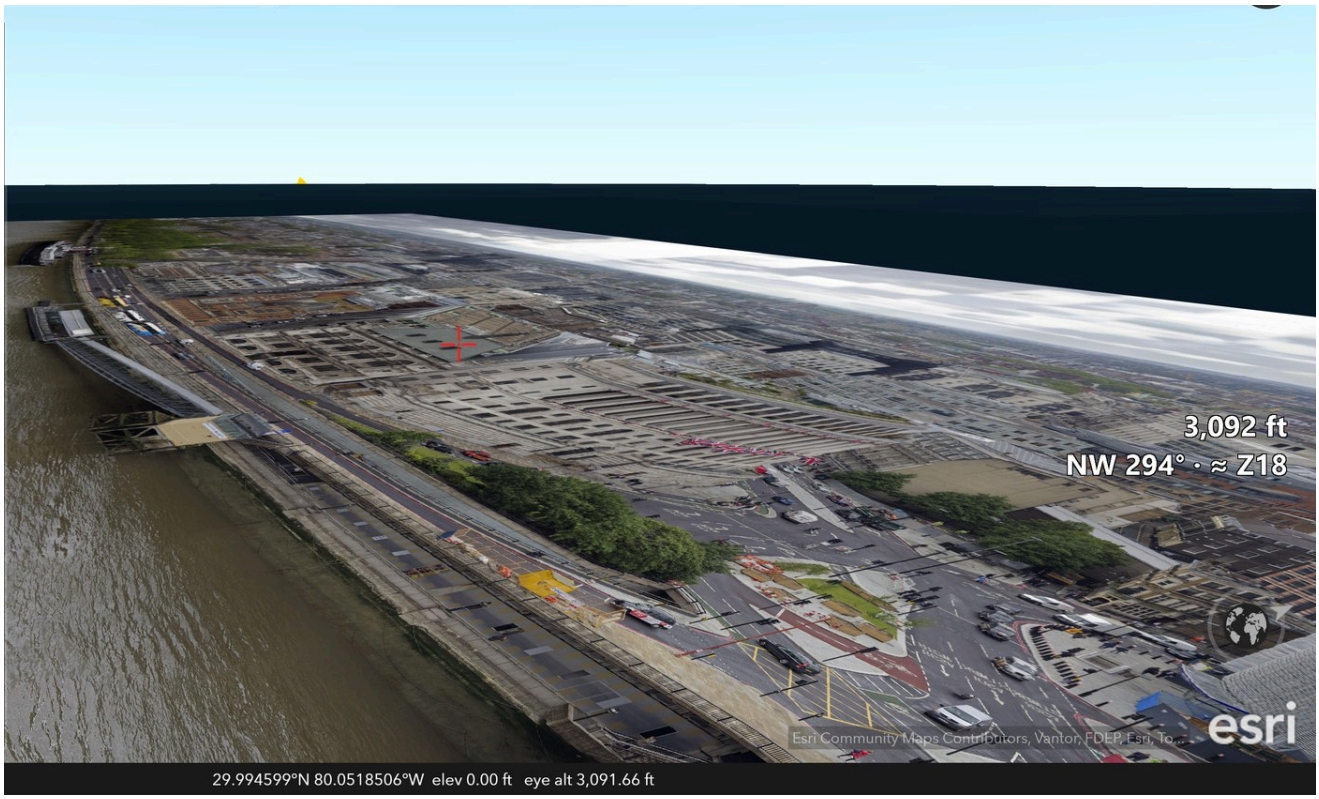
1. **Extract All first:** right-click the downloaded zip > **Extract All**, then open the new folder. (Double-clicking files while they are still inside the zip does not work.)
2. **Double-click** `Start Picture World Maker.bat` - the only .bat in the folder. This manual sits right next to it. The **System Files** folder holds the program - leave it alone.
3. **Blue "Windows protected your PC" box?** Click **More info**, then **Run anyway**. It only asks the first time. (Or, before extracting: right-click the zip > **Properties** > tick **Unblock** > **OK**.)
4. **Python - nothing to install by hand.** If Python is missing, the window asks **Install Python now? [Y/N]**: type **Y** and press Enter. It installs just for you (no admin needed) - and the Pillow picture library too. On an older PC this can take several minutes and Windows Defender may make the PC look frozen - **do not close the window**.
5. To fly over your picture you need **ArcGIS Earth** (free from Esri: https://www.esri.com/en-us/arcgis/products/arcgis-earth/downloads).
6. Giant pictures need room: free disk space of a few times the picture's size, and plenty of memory.

Something stops it? The window says what to do in plain words, and keeps a log in `System Files\Ready_To_Fly_log.txt`. On a work computer that blocks installs, ask IT to install Python 3 from python.org.

*Formerly Rasta Pyramid Factory.*

Printable copy: Picture_World_Maker_User_Manual.pdf

**Slice. Dice. Present.** Picture World Maker turns a giant picture - a city panorama, a drone map, a scanned map, artwork - into a zoomable map you can open in **ArcGIS Earth**, then fly over and dive into. *See the whole scene like a hawk. Dive into any detail that catches your eye.*




*A London panorama photo (50,000 × 12,006 pixels) made with Picture World Maker v0.1.6 and flown in ArcGIS Earth. A photo has no real location, so Picture World Maker parks it on the ocean off Jacksonville, Florida: the status bar shows 29.99 N, 80.05 W, elevation 0 ft. Every spot in the picture now has its own latitude and longitude, like a real map.*

## What you need

- **Windows 10 or 11** (64-bit).
- **Python 3** (3.9 or newer) and **Pillow** - free, and `Start Picture World Maker.bat` installs both for you if they are missing (answer **Y**; no admin needed). Or get Python from python.org.
- **Pillow** - a free Python picture helper; **Start Picture World Maker.bat installs it by itself** the first time.
- **ArcGIS Earth** - free, to look at and fly over the result.
- **Memory:** about 5 bytes per picture pixel - about 3 GB for a 600-million-pixel photo, about 6 GB for 1.2 billion.
- **Free disk space on C:** - room for the finished file twice while it builds (the working files go in the Windows temporary folder and are deleted afterwards). Picture World Maker checks this before it starts and tells you if there is not enough room.

## Install

1. Download `Picture_World_Maker_v2.1.zip`, right-click it > **Extract All**.

2. Open the `Picture_World_Maker_v2.1` folder. It holds just three things: **Start Picture World Maker.bat** (the one to double-click), this manual as a PDF, and the **System Files** folder (leave it alone).

## What's in the zip




## Build a picture map

1. Double-click **Start Picture World Maker.bat**.
2. **1. Choose giant raster** - press **CHOOSE IMAGE** and pick your picture (JPG, PNG, TIF, BMP or WEBP). Picture World Maker checks it and shows its size and how many tiles it will make. *Phone photos are turned the right way up, 16-bit pictures (scans, space pictures) are brightened to normal, and see-through parts of a PNG come out white (2.1).*
3. **2. Pyramid recipe** - leave **Min zoom 0** and **Max zoom 18** for a normal picture. Picture World Maker suggests Max zoom 18: that is where you reach the photo's own detail. **World spot** - which spot on the ocean the picture is parked at (1 to 20, see World spots). Leave it on **1** for a single picture; give each picture of a gallery its own spot.
4. **3. Finished product** - **TPKX** for ArcGIS Earth, **MBTiles** for other map programs, or **Both**. Type a **Name** and choose a **Folder** (it starts on your Desktop). The Name box fills in from the first picture you choose - when you build a second picture, type its name yourself. The file name gets its World spot added at the end, for example `Tower_of_Babel_spot02.tpkx`.
5. Press **BUILD RASTER PYRAMID** and leave it running. The status line keeps moving - a heartbeat, the time, and the stage (building the pyramid, converting to TPKX, verifying).
6. When it says **RASTER PYRAMID COMPLETE**, your file is in the folder you chose.




How long? About **1 minute** (55 s on Jim's PC) for a 50,000 × 12,006 picture (35,216 tiles, TPKX 442 MB).

**Stopping:** press **CANCEL** - Picture World Maker stops within a second or two, cleans up and says *Build cancelled*. Closing the window with the X does the same before it closes.

## Look at it in ArcGIS Earth

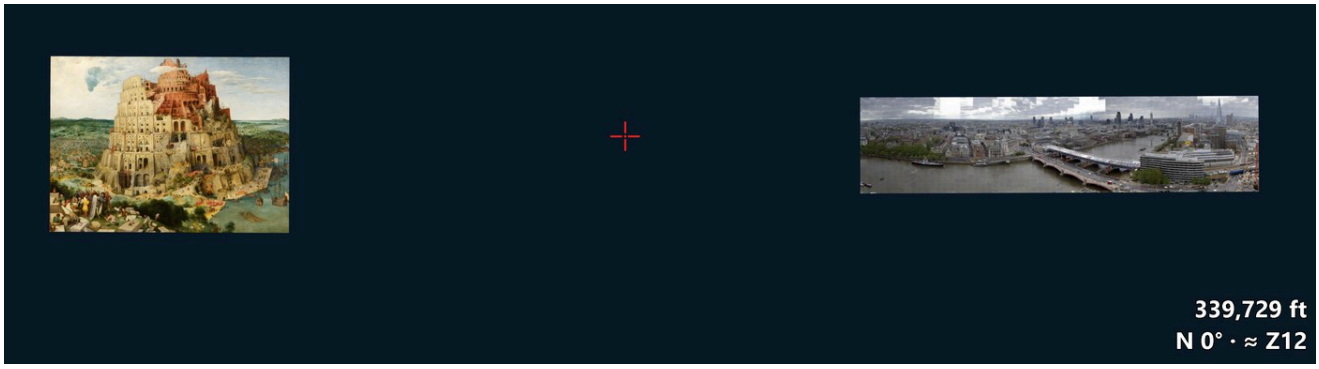
1. Drag the `.tpkx` file onto the ArcGIS Earth map (or use the **+** Add data button).
2. If the view does not jump there, right-click the layer in the list and zoom to it.
3. **Where is it?** A picture has no location, so Picture World Maker places it on the **Atlantic Ocean east of Jacksonville**, centred on its **World spot** (spot 1 = 30 N, 80 W), at **1 picture pixel = 1 map metre**. The ocean is flat and empty, so the picture lies there like a giant carpet.
4. Zoom in: the detail sharpens down to the photo's own pixels.

## World spots: a gallery on the ocean

Since v0.2.4 every picture can have its **own place on the ocean**. Choose a **World spot** in the window before you build. Each spot is the centre of its own one-degree box of open Atlantic east of Florida, about 111,000 × 128,000 picture pixels at 1 pixel per metre, so even very large pictures fit without touching their neighbours. If a picture is bigger than its box, the window says so and asks before building.

- **Spot 1** is where every picture landed before v0.2.4, so your older files stay where they are.

- Spots 2-10 step one degree east along 30 N; spots 11-20 are the row one degree north, at 31 N.
- All 20 spots are open ocean, well clear of the coast, the Bahamas and Bermuda.
- Turn off a picture's layer in ArcGIS Earth when you build a different picture on the same spot - two pictures on one spot lie on top of each other.



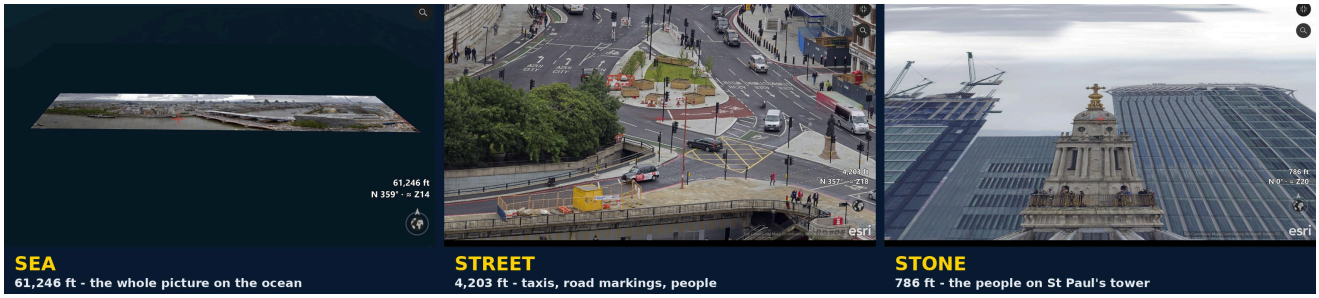

*Live-proven 2026-11-06: Bruegel's* Tower of Babel *(public domain, 30,000 × 21,952 px) on World spot 1 and the London panorama on World spot 2 (built in 53 s, 35,205 tiles), side by side on the Atlantic. Jim: "Of all the crazy things to find floating in the Atlantic Ocean!!!"*

**Gates to your pictures:** in World Gate Maker, use a spot's coordinates below with **ground elev 0** to make a gate pair from home to any picture. One gate per picture, each leading to the next, makes a **gallery tour** - an artist's paintings one after another.

| World spot | Latitude | Longitude |
|---|---|---|
| 1 | 30.0 | -80.0 |
| 2 | 30.0 | -79.0 |
| 3 | 30.0 | -78.0 |
| 4 | 30.0 | -77.0 |
| 5 | 30.0 | -76.0 |
| 6 | 30.0 | -75.0 |
| 7 | 30.0 | -74.0 |
| 8 | 30.0 | -73.0 |
| 9 | 30.0 | -72.0 |
| 10 | 30.0 | -71.0 |
| 11 | 31.0 | -80.0 |
| 12 | 31.0 | -79.0 |
| 13 | 31.0 | -78.0 |
| 14 | 31.0 | -77.0 |
| 15 | 31.0 | -76.0 |
| 16 | 31.0 | -75.0 |
| 17 | 31.0 | -74.0 |
| 18 | 31.0 | -73.0 |
| 19 | 31.0 | -72.0 |
| 20 | 31.0 | -71.0 |

## Fly into it with EarthStick

With EarthStick you can fly over the picture, hover above any part, and dive in. The readout's **≈ Z** number tells you when you have reached the sharpest detail (about Z18).




*Sea, street, stone - one photograph, three heights: the whole panorama floating on the ocean, the street with its taxis and road markings, and the people on St Paul's bell tower.*

**Gates into your picture:** because every spot of the picture has a latitude and longitude (hover the mouse over it and read the ArcGIS Earth status bar), you can make World Gates that drop you right onto a landmark in the photo. In World Gate Maker, type those coordinates with **ground elev 0** and a height such as 300 ft.

The picture always lies flat - it is a map layer, so it cannot stand up like a billboard. A photo taken at an angle keeps its own perspective, which gives a strange, half-3D diorama feel when you fly over it.

## Messages and fixes

- **"Not enough memory to open this picture"** - close other programs and try again, or use a smaller picture.
- **Want all-PNG tiles (no JPEG)?** - set `tiles = png` under `[PHOTO]`. Bigger files, no JPEG compression.
- **"has map coordinates"** - the file is a real map (a GeoTIFF). ArcGIS Earth can open it directly: just drag it on. Picture World Maker still builds it, as a picture parked on the ocean.
- **"TPKX supports zoom levels up to 23"** - TPKX (Esri's tile packages) stop at zoom 23. Lower Max zoom, or choose MBTiles only (up to 25).
- **"... already exists. Replace it?"** - a file with that name is already in the folder. **Yes** builds over it; **No** keeps the old file and builds nothing (then pick another Name).
- **"Not enough disk space"** - the drive is too full for this picture. Free some space, choose a lower Max zoom, or save to another drive.
- **It runs for a long time** - giant pictures take minutes; the moving heartbeat shows it is working. The size warning before the build tells you how many tiles to expect.

## What Picture World Maker does not do

- It cannot add detail that is not in the photo - a blurry photo stays blurry when you zoom in.
- A photo parked on the ocean is not real geography; its location is a display spot, chosen so the map tools can work.
- The original picture is never changed.